HYPERHEURISTIC OPTIMIZATION OF UNIVERSITY COURSE TIMETABLING PROBLEM USING REINFORCEMENT LEARNING AND GENETIC ALGORITHM

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Presentation Outline

- Motivation
- Introduction
- Objectives
- Project Applications
- Scope Of Project
- Methodology
- Result
- Analysis
- Future Enhancements
- Conclusion
- References

Motivation

- Motivation point 1: Describe the real-world problem or need
- Motivation point 2: Why this problem matters
- Motivation point 3: Current gaps or limitations
- Motivation point 4: What inspired this project

Introduction

- Brief background of the project domain
- Key concepts and context
- Overview of the proposed system/solution

Objectives

Note: Place the exact objectives as you submitted in the report. Remove this box before presenting.

- Objective 1: Primary goal of the project
- Objective 2: Secondary goal
- Objective 3: Additional goal
- Objective 4: Extended goal (if applicable)

Literature Review

Paper	Author	Objective	Methodology	Findings	Limitations
Smart Traffic Signal Control (2021)	Sharma et al.	Reduce urban congestion using adaptive signals	Reinforcement learning with Q-learning	30% reduction in avg. wait time	Tested only on simulated data
Crop Disease Detection (2022)	Patel & Kumar	Identify plant diseases from leaf images	CNN with transfer learning (ResNet-50)	94.5% accuracy on test set	Limited to 5 crop species
IoT-Based Air Quality (2023)	Adhikari et al.	Real-time air quality monitoring system	Sensor network with cloud dashboard	Detected pollutant spikes within 2 min	High sensor calibration cost

Scope Of Project

- Scope point 1: Define the boundaries and coverage of the project
- Scope point 2: What the project will deliver and its limitations

Project Applications

- Application 1: Primary use case of the project
- Application 2: Secondary practical application
- Application 3: Broader impact or accessibility benefit

Methodology - [1] (System Block Diagram)

[System Block Diagram]

Insert your system block diagram here

Methodology - [2] (Working Principle)

- Working principle point 1: How the core system operates
- Working principle point 2: Sensor/data coordination
- Working principle point 3: Data capture mechanism
- Working principle point 4: Storage approach
- Working principle point 5: Data transmission and analysis

Methodology - [3] (Operational Workflow)

- Workflow step 1: Signal preprocessing and filtering
- Workflow step 2: Noise/artifact handling
- Workflow step 3: Feature extraction from processed data
- Workflow step 4: Classification or decision-making
- Workflow step 5: Aggregation and final output

Methodology - [4] (Flow Of Proposed System)

[System Flowchart]

Insert your proposed system flowchart here

Methodology - [5] (Hardware Requirement)

- Hardware component 1 (e.g. Microcontroller)
- Hardware component 2 (e.g. Sensor module)
- Hardware component 3 (e.g. Motion sensor)
- Hardware component 4 (e.g. Battery)
- Hardware component 5 (e.g. Power converter)

Methodology - [6] (Software Requirement)

- Software tool 1 (e.g. IDE)
- Software tool 2 (e.g. Programming language)
- Software tool 3 (e.g. Web framework)
- Software tool 4 (e.g. ML library)
- Software tool 5 (e.g. Version control)
- Software tool 6 (e.g. Simulation tool)

Methodology - [7] (Machine Learning Architecture)

- Description of the ML technique used
- Training approach and data splitting strategy
- How prediction/classification is performed
- Measures to prevent overfitting and improve generalization

Methodology - [8] (Algorithm Architecture)

[Algorithm Architecture Diagram]

Insert your algorithm architecture diagram here

Methodology - [9] (Dataset Preparation)

- Data source and collection method
- Key variables and measurements
- Labeling and annotation approach
- Feature extraction from raw data
- Train/validation/test split strategy

Methodology - [10] (Snapshot of Dataset)

[Dataset Snapshot]

Insert a sample table or screenshot of your dataset here

Methodology - [11] (Model Training)

- Model/classifier training details
- Class balancing or weighting strategy
- Hyperparameter tuning and optimization metric
- Evaluation on held-out test set
- Model serialization and deployment approach

Expected Outcome

- Expected outcome 1: What the system is expected to achieve
- Expected outcome 2: Anticipated performance or capability
- Expected outcome 3: Deliverables upon completion
- Expected outcome 4: Measurable success criteria

Future Enhancements

- Future enhancement 1: Broader dataset and improved generalization
- Future enhancement 2: Advanced model architecture for better performance
- Future enhancement 3: Edge deployment for real-time operation

Conclusion

- Conclusion point 1: Summary of the proposed/developed system
- Conclusion point 2: Key contributions and significance
- Conclusion point 3: Feasibility and approach taken
- Conclusion point 4: Results achieved or expected outcomes
- Conclusion point 5: Practical impact and relevance

References

- Reference 1: Author(s), "Title of paper/article," Journal/Conference, Date. Available: URL. [Accessed: Date]
- Reference 2: Author(s), "Title of paper/article," Journal/Conference, vol. X, no. Y, pp. XX–YY, Year

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- Reference 3: Author(s), “Title of paper/article,” Journal/Conference, vol. X, p. XXXXX, Month. Year
- Reference 4: Author(s), “Title of paper/article,” Journal/Conference, vol. X, no. Y, pp. XX–YY, Year